

AutoNetKit User Manual

Type-Safe Network Topology Modelling

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About this manual. This manual covers installing AutoNetKit, building typed network topologies with Pydantic models, deriving multi-layer protocol views, querying topology data with the fluent API, running design-rule validation before configuration generation, and compiling validated designs into vendor-specific device configurations. Chapters on the layer system, the query DSL, built-in design rules, supported output platforms, and ready-made batteries_included topologies make this the primary practitioner reference for AutoNetKit 2.1.0.

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1 Quick Start

Go from zero to a running, validated topology in under ten minutes.

Step 1. Install AutoNetKit (the NTE Rust engine installs automatically):

```
pip install autonetkit
```

Step 2. Create and populate a topology:

```
from pydantic import BaseModel
from autonetkit import Topology
from autonetkit.models.base import BaseTopologyNode, BaseTopologyEndpoint
from autonetkit.models.edge import BaseInternodeEdge

class RouterData(BaseModel):
    label: str
    asn: int = 65000
    platform: str = "cisco_ios"

class Router(BaseTopologyNode[RouterData]):
    pass

class InterfaceData(BaseModel):
    label: str

class Interface(BaseTopologyEndpoint[InterfaceData]):
    pass

class LinkData(BaseModel):
    bandwidth: int | None = None

class Link(BaseInternodeEdge[LinkData]):
    type: str = "link"
```

Step 3. Build the physical topology:

```
topo = Topology()
r1 = Router(layer="physical", data=RouterData(label="R1"))
r2 = Router(layer="physical", data=RouterData(label="R2"))
topo.nodes.add([r1, r2])

# Add interfaces and connect them
i1 = topo.endpoint_mgr.add(
    Interface, layer="physical",
    data=InterfaceData(label="Eth0/0"), parent_id=r1.id,
)
i2 = topo.endpoint_mgr.add(
    Interface, layer="physical",
    data=InterfaceData(label="Eth0/0"), parent_id=r2.id,
)
topo.edges.add(Link(data=LinkData(bandwidth=10000), src=i1, dst=i2))
```

Step 4. Derive protocol layers, validate, and compile:

```
# Derive an OSPF layer
```

```

topo.layers.build({"ospf": {"parent": "physical",
                           "keep_edges_where_same": "asn"}})

# Validate the design
from autonetkit.analysis import standard_ruleset
report = standard_ruleset().run(topo)
print(report.to_text())

# Compile to IOS configuration
from autonetkit.compilers.registry import get_compiler
compiler = get_compiler("ios")
config = compiler.compile(topo, r1.id)
print(compiler.render_config(config))

```

Step 5. Query node counts per layer:

```

# In a Python REPL or script
for name in topo.layers.names():
    print(name, topo.query.nodes().in_layer(name).count())

```

2 Installation and Prerequisites

2.1 Requirements

▷ Prerequisites

- **Python 3.9 or later** (3.11+ recommended for best performance).
- **Pydantic v2** — installed automatically as a dependency.
- **NTE engine** (`ank_nte`) — the Rust graph-storage backend; installed automatically as a dependency.
- **Polars** — columnar query execution; installed automatically.
- An active Python virtual environment is strongly recommended.

2.2 Installing from PyPI

```
pip install autonetkit
```

Tip

Use `uv pip install autonetkit` for significantly faster dependency resolution in projects with large dependency trees. `uv` resolves the Rust wheel for your platform in a single round-trip rather than iterating through `pip`'s backtracking solver.

Caution

The NTE engine (`ank_nte`) ships pre-built wheels for Linux (x86_64, aarch64), macOS (Intel, Apple Silicon), and Windows (x86_64). If you are on an unsupported platform, `pip` will attempt a source build. This requires a stable Rust toolchain (`rustup`) and the `maturin` build tool. Install Rust first with:

```
curl -proto '=https' -tlsv1.2 -sSf https://sh.rustup.rs | sh
```

2.3 Verifying the Installation

```
python -c "import autonetkit; print(autonetkit.__version__)"
python -c "import ank_nte; print(ank_nte.__version__)"
```

Both commands should print `2.1.0` and the NTE version string respectively.

2.4 Upgrading

```
pip install --upgrade autonetkit
```

Caution

AutoNetKit 2.x uses a different NDJSON serialisation protocol to 1.x. Topology objects serialised under 1.x cannot be loaded by 2.x directly. Use `autonetkit.io.migrate_v1(path)` to convert legacy NDJSON exports before loading them into a 2.x Topology.

3 Core Concepts

AutoNetKit organises a network topology into three kinds of element:

- Nodes** represent devices or logical network elements: routers, switches, hosts, VLANs, VRFs. Nodes are first-class graph citizens.
- Endpoints** represent connection points on devices: physical interfaces, loopbacks, sub-interfaces. Each endpoint belongs to exactly one node via an ownership edge.
- Edges** represent relationships between elements. Three edge kinds enforce strict connection rules: *internode* (endpoint to endpoint across devices), *intranode* (node to node within a device), and *ownership* (any node to any node, containment).

Every element carries a generic data: `T` field backed by a user-defined Pydantic `BaseModel`. This gives full IDE autocompletion, runtime validation, and free serialisation — with no stringly-typed dictionaries.

The Topology façade exposes managers for every domain:

Command Reference

Command	Description
<code>topology.nodes</code>	Add, remove, and register node models
<code>topology.endpoint_mgr</code>	Add and query endpoints (interfaces)
<code>topology.edges</code>	Add, remove, and query edges
<code>topology.links</code>	Simplified link-level (node-to-node) view
<code>topology.layers</code>	Create, derive, freeze, and query layers
<code>topology.query</code>	Fluent query API over nodes, edges, links, paths
<code>topology.io</code>	Serialise / deserialise (NDJSON, JSON, GraphML)
<code>topology.analysis</code>	Run rule sets and retrieve reports

The Topology class is a thin façade wiring approximately eighteen domain managers over a shared NTE store. Use dot-completion to explore the API.

4 Core Workflows

4.1 Defining Custom Models

Every domain-specific element type inherits from one of AutoNetKit’s base classes parameterised by a data model:

Step 1. Define a Pydantic data class for each element kind:

```
from pydantic import BaseModel

class RouterData(BaseModel):
    label: str
```

Inherit `BaseTopologyNode` for devices and `BaseTopologyEndpoint` for interfaces. Both are generic over a Pydantic `BaseModel` subclass.

```

    role: str = "core"          # "core" | "edge" | "spine" | "leaf"
    asn: int = 65000
    platform: str = "cisco_ios"
    loopback: str | None = None

class InterfaceData(BaseModel):
    label: str
    speed: str | None = None    # e.g. "10G", "100G"
    ip: str | None = None

class LinkData(BaseModel):
    bandwidth: int | None = None # Mbps
    metric: int = 1

```

Step 2. Inherit the appropriate base class, parameterised by your data model:

```

from autonetkit.models.base import (
    BaseTopologyNode, BaseTopologyEndpoint,
)
from autonetkit.models.edge import BaseInternodeEdge

class Router(BaseTopologyNode[RouterData]):
    pass # inherits id, layer, data; extend for domain logic

class Interface(BaseTopologyEndpoint[InterfaceData]):
    pass

class Link(BaseInternodeEdge[LinkData]):
    type: str = "link" # discriminator used by registry

```

Step 3. Register models with the topology so the identity map can hydrate them:

```

topo = Topology()
topo.nodes.register_models([Router, Interface])
topo.edges.register_models([Link])

```

Tip

Use `FlexibleData` (from `autonetkit.models.base`) as the data type during rapid prototyping. It sets `extra="allow"` so any field is accepted without a formal schema. Migrate to a typed model before running design rules or generating configurations.

4.2 Building the Physical Topology

Step 1. Instantiate a Topology and add nodes:

```

from autonetkit import Topology

topo = Topology()

core1 = Router(layer="physical",
               data=RouterData(label="Core1", role="core"))
core2 = Router(layer="physical",
               data=RouterData(label="Core2", role="core"))
topo.nodes.add([core1, core2])
# IDs are assigned by NTE after add(); router.id is now set

```

Step 2. Add endpoints (interfaces) owned by each node:

```
# endpoint_mgr.add() creates the endpoint and the ownership edge
c1_eth0 = topo.endpoint_mgr.add(
    Interface, layer="physical",
    data=InterfaceData(label="Eth0/0", speed="10G"),
    parent_id=core1.id,
)
c2_eth0 = topo.endpoint_mgr.add(
    Interface, layer="physical",
    data=InterfaceData(label="Eth0/0", speed="10G"),
    parent_id=core2.id,
)
```

Step 3. Connect endpoints with an internode edge (physical link):

```
link = Link(
    data=LinkData(bandwidth=10000, metric=1),
    src=c1_eth0,
    dst=c2_eth0,
)
topo.edges.add(link)
```

Step 4. Verify the topology:

```
print(topo.query.nodes().in_layer("physical").count()) # 2
print(topo.query.edges().in_layer("physical").count()) # 1
```

Caution

Node IDs are assigned by NTE *after* calling `topo.nodes.add()`. Do not capture `node.id` before the node has been added — it will be `None`. Batch-add is more efficient than adding nodes one by one for topologies with more than a few hundred elements.

4.3 Deriving Protocol Layers

Protocol views are derived from the physical layer using a configuration dictionary:

Step 1. Define the layer configuration:

```
LAYERS = {
    "ospf": {
        "parent": "physical",
        # Clone edges only where both endpoints share the same ASN
        "keep_edges_where_same": "asn",
    },
    "ebgp": {
        "parent": "physical",
        # Clone edges only where endpoints cross AS boundaries
        "keep_edges_where_different": "asn",
    },
    "ibgp": {
        "parent": "physical",
        "copy_edges": False,
        # Group nodes by ASN; create a full mesh within each group
        "group_by": "asn",
        "create": "full_mesh",
    },
}
```

Layer derivation is declarative, not reactive. Call `layers.build()` after all physical nodes and links are in place.

Step 2. Validate and build:

```
errors = topo.layers.validate(LAYERS)
if errors:
    for e in errors:
        print("Layer config error:", e)
else:
    topo.layers.build(LAYERS)
```

Step 3. Optionally freeze source layers to prevent accidental mutation:

```
topo.layers.freeze("physical")
assert topo.layers.frozen("physical")
```

4.4 Querying Topology Data

AutoNetKit's query API is fluent and type-safe:

Step 1. Query all nodes of a type in a layer:

```
# List[Router] -- fully hydrated Pydantic models
core_routers = (
    topo.query.nodes()
    .of_type(Router)
    .in_layer("physical")
    .where(role="core")
    .models()
)
```

Queries are lazy builders. Terminal methods (.models(), .ids(), .count()) execute against Rust.

Step 2. Use expression filters for complex predicates:

```
from autonetkit import q

# Field comparison
high_bw_links = (
    topo.query.links()
    .in_layer("physical")
    .filter(q.field("bandwidth") >= 100000)
    .collect()
)

# String match + logical AND
cisco_core = (
    topo.query.nodes()
    .of_type(Router)
    .filter(
        (q.field("platform").contains("cisco")) &
        (q.field("role") == "core")
    )
    .models()
)
```

Step 3. Use terminal methods suited to the task:

```
count = topo.query.nodes().of_type(Router).count()
exists = topo.query.nodes().where(role="spine").exists()
```

```
first = topo.query.nodes().of_type(Router).first()
one   = topo.query.nodes().where(label="Core1").one()
df    = topo.query.nodes().of_type(Router).dataframe()
```

Step 4. Navigate across layers:

```
# Find the plan-layer counterpart of a physical router
plan_router = physical_router.in_plan(topo)

# Find the same router in the OSPF layer
ospf_router = physical_router.in_layer("ospf", topo)
```

4.5 Running Design Rules

Step 1. Run the built-in standard ruleset:

```
from autonetkit.analysis import standard_ruleset

report = standard_ruleset().run(topo)
print(report.to_text(verbose=True))
```

Step 2. Check the result and act on failures:

```
if not report.passed:
    print(f"Errors: {report.error_count}, "
          f"Warnings: {report.warning_count}")
    for r in report.filter_failed():
        print(f" [{r.severity.value.upper()}] {r.message}")
    raise SystemExit(1)
```

Step 3. Export the report as a Polars DataFrame for CI integration:

```
df = report.to_dataframe()
df.write_csv("validation_report.csv")
```

4.6 Generating Device Configurations

Step 1. Look up the compiler for a platform:

```
from autonetkit.compilers.registry import get_compiler

compiler = get_compiler("ios")          # Cisco IOS
# or: get_compiler("eos"), get_compiler("junos"), ...
```

Step 2. Compile and render for each device:

```
for router in topo.query.nodes().of_type(Router).models():
    compiler = get_compiler(router.data.platform)
    config = compiler.compile(topo, router.id)
    cli_text = compiler.render_config(config)
    out_path = f"output/{router.data.label}.cfg"
    with open(out_path, "w") as f:
        f.write(cli_text)
```

Step 3. Export for a deployment environment:

```

from autonetkit.export import ContainerlabExporter

exporter = ContainerlabExporter()
exporter.export(topo, output_dir="lab/")
# Produces lab/topology.clab.yml and per-device config files

```

5 Layer Configuration Reference

A layer configuration dictionary maps layer names to parameter objects. The build system resolves dependencies, validates the configuration, and builds layers in topological order.

5.1 Parameters

Layer Parameters

Command	Description
parent	Name of the parent layer to clone nodes from
keep_edges_where_same	Clone edges only where data.<field> matches on both ends
keep_edges_where_different	Clone edges only where data.<field> differs
copy_edges	Boolean; if False, no edges are copied (default: True)
group_by	Field name; groups nodes by field value for topology creation
create	Topology pattern within each group: full_mesh, hub_spoke, ring

5.2 Common Patterns

5.2.1 Full Mesh (iBGP)

Nodes grouped by ASN; a full mesh of virtual links is created within each group. This models iBGP sessions without requiring explicit physical links.

```

ibgp:
  parent: physical
  copy_edges: false
  group_by: asn
  create: full_mesh

```

5.2.2 Same-AS OSPF

Only physical links between routers in the same AS are retained.

```

ospf:
  parent: physical
  keep_edges_where_same: asn

```

5.2.3 Cross-AS eBGP

Only physical links crossing AS boundaries are retained.

```

ebgp:
  parent: physical
  keep_edges_where_different: asn

```

5.2.4 Ring Topology

Group nodes by a field and connect them in a ring (useful for modelling ring access networks).

```
access_ring:
  parent: physical
  copy_edges: false
  group_by: region
  create: ring
```

5.2.5 Hub and Spoke

Group nodes by a field; the node with the highest degree within each group becomes the hub.

```
hub_spoke:
  parent: physical
  copy_edges: false
  group_by: pop
  create: hub_spoke
```

5.3 Cross-Layer Traversal

After derivation, every cloned node carries pointers back to its parent:

```
# Get the physical counterpart of an OSPF node
ospf_router = topo.query.nodes().of_type(Router) \
    .in_layer("ospf").first()

physical_router = ospf_router.in_plan(topo)      # -> plan layer
ibgp_router      = ospf_router.in_layer("ibgp", topo) # -> iBGP layer
```

6 Python API Quick Reference

AutoNetKit is primarily a library. The following tables summarise the most commonly used manager methods.

6.1 Node and Endpoint Management

NodeManager — `topology.nodes`

Command	Description
<code>.add(nodes)</code>	Add a list of nodes; assigns IDs in place
<code>.remove(node_id)</code>	Remove node and all its edges
<code>.get(node_id)</code>	Retrieve a node by ID
<code>.register_models(types)</code>	Register model classes with the identity map
<code>.count()</code>	Total node count across all layers

EndpointManager — `topology.endpoint_mgr`

Command	Description
<code>.add(cls, layer, data, parent_id)</code>	Create endpoint and ownership edge
<code>.get(ep_id)</code>	Retrieve endpoint by ID
<code>.of_node(node_id)</code>	List all endpoints owned by a node

6.2 Edge and Link Management

EdgeManager — `topology.edges`

Command	Description
<code>.add(edge)</code>	Add a typed edge; assigns ID in place
<code>.remove(edge_id)</code>	Remove edge
<code>.get(edge_id)</code>	Retrieve edge by ID
<code>.count()</code>	Total edge count

LinkManager — `topology.links`

Command	Description
<code>.between(a_id, b_id)</code>	Find links between two nodes
<code>.neighbours(node_id)</code>	List neighbour node IDs
<code>.degree(node_id)</code>	Connection degree of a node

6.3 Layer Management

LayerManager — `topology.layers`

Command	Description
<code>.build(config)</code>	Build layers from config dict
<code>.validate(config)</code>	Validate config dict; returns error list
<code>.get(name)</code>	Return a LayerHandle
<code>.names()</code>	List all layer names
<code>.freeze(name)</code>	Prevent further mutation
<code>.frozen(name)</code>	Check frozen status
<code>.clone_to_layer(...)</code>	Clone a node set to a new layer

6.4 Query API

QueryManager — `topology.query`

Command	Description
<code>.nodes()</code>	Start a NodeQuery
<code>.edges()</code>	Start an EdgeQuery
<code>.links()</code>	Start a LinkQuery
<code>.paths()</code>	Start a PathQuery

NodeQuery chaining

Command	Description
<code>.of_type(Model)</code>	Type-narrowing filter
<code>.in_layer(name)</code>	Layer filter
<code>.where(**fields)</code>	Equality filter on <code>data.*</code> fields
<code>.filter(*exprs)</code>	Expression-based filter using <code>q.*</code>
<code>.to_plan()</code>	Navigate to plan-layer counterparts
<code>.to_layer(name)</code>	Navigate to another layer

NodeQuery terminals

Command	Description
<code>.models()</code>	<code>List[T]</code> of hydrated Pydantic models
<code>.ids()</code>	<code>List[int]</code> of node IDs
<code>.count()</code>	<code>int</code> (no model materialisation)
<code>.first()</code>	<code>Optional[T]</code> — first match or <code>None</code>
<code>.one()</code>	<code>T</code> — exactly one match (raises if 0 or 2+)
<code>.exists()</code>	<code>bool</code>
<code>.dataframe()</code>	<code>pl.DataFrame</code> (Polars)

6.5 IO and Serialisation

IOManager — `topology.io`

Command	Description
<code>.save(path)</code>	Serialise topology to NDJSON
<code>.load(path)</code>	Deserialise from NDJSON
<code>.to_json()</code>	Return JSON string
<code>.to_graphml(path)</code>	Export to GraphML (for visualisation)
<code>.migrate_v1(path)</code>	Convert 1.x NDJSON to 2.x format

7 Design Rules Catalogue

The rule engine is AutoNetKit’s “network linter”: composable, declarative validation that runs before configuration generation, catching design errors before they become operational incidents.

7.1 Built-In Rules

AutoNetKit ships 27 built-in rules across five categories. The `standard_ruleset()` factory runs the four most critical.

Category	Rule ID	Purpose	Severity
Validation	validation.isolated_nodes	Nodes with no connections	ERROR
	validation.required_field	Mandatory field is set	ERROR
	validation.min_connections	Minimum degree per node type	ERROR
	validation.unique_labels	No duplicate node labels	WARNING
	validation.no_self_loops	Edges do not connect a node to itself	ERROR
Anomaly	anomaly.spoof	Articulation points (single PoF)	WARNING
	anomaly.disconnected	Partitioned topology components	ERROR
	anomaly.high_degree	Nodes above degree threshold	WARNING
	anomaly.orphaned_endpoint	Endpoints not connected to any link	WARNING
	anomaly.stub_asn	ASN with only one external link	INFO
	anomaly.duplicate_link	Parallel links between same pair	INFO
Pattern	pattern.hub_spoke	Star topology detected	INFO
	pattern.full_mesh	Fully connected clique	INFO
	pattern.ring	Ring topology detected	INFO
	pattern.asymmetric_degree	Unbalanced degree distribution	WARNING
	pattern.triangle_free	No triangles in physical layer	INFO
Resilience	resilience.dual_homed	Nodes with ≥ 2 upstreams	WARNING
	resilience.path_diversity	Edge-disjoint path count	WARNING
	resilience.link_redundancy	Redundant links per pair	INFO
	resilience.min_upstreams	Minimum upstream count	ERROR
	resilience.max_radius	Diameter bound	WARNING
	resilience.ecmp_capacity	Equal-cost path count	INFO
Operational	operational.missing_attr	Required attribute absent	ERROR
	operational.asymmetric_link	Speed mismatch across a link	WARNING
	operational.platform_support	Platform has a registered compiler	ERROR
	operational.loopback_missing	Routers without a loopback address	WARNING
	operational.asn_conflict	Same ASN used by unrelated routers	ERROR

7.2 Rule Composition Operators

Five operators build complex rules from simple leaf rules:

```

from autonetkit.analysis import AllOf, AnyOf, IfThen, Not, AtLeast

# All listed rules must pass
mandatory = AllOf(
    IsolatedNodesRule(),
    MinConnectionsRule(Router, min_count=2),
    rule_id="mandatory_checks",
)

# Conditional: only check dual-homing on core routers
core_check = IfThen(
    condition=NodeTypeRule(CoreRouter),
    consequence=DualHomedRule(CoreRouter, min_upstreams=3),
    rule_id="core_redundancy",
)

# At least 2 of 3 resilience rules must pass
resilience = AtLeast(2,
    PathDiversityRule(min_paths=2),
    LinkRedundancyRule(),

```

```
DualHomedRule(Router, min_upstreams=2),
rule_id="resilience_floor",
)
```

7.3 Writing a Custom Rule

```
from autonetkit.analysis.base import Rule, RuleResult, Severity

class MinInterfacesRule(Rule):
    id = "custom.min_interfaces"
    name = "Minimum Interfaces"
    description = "Each router must have at least N interfaces."
    tags = {"validation", "connectivity"}
    default_severity = Severity.ERROR

    def __init__(self, min_count: int = 2):
        self.min_count = min_count

    def check(self, ctx) -> list[RuleResult]:
        results = []
        for router in ctx.nodes_of_type(Router):
            degree = ctx.get_degree(router.id)
            results.append(RuleResult(
                rule_id = self.id,
                passed = degree >= self.min_count,
                severity = self.default_severity,
                message = (f"{router.data.label} has {degree} "
                           f"interfaces (minimum: {self.min_count})"),
                affected_nodes = [router.id] if degree < self.min_count else [],
            ))
        return results

# Register and run
from autonetkit.analysis import RuleSet
ruleset = RuleSet("custom_checks")
ruleset.add(MinInterfacesRule(min_count=2))
report = ruleset.run(topo)
```

Tip

Tag rules with category strings ("validation", "resilience", etc.) and use `report.results_by_tag()` to group the report output by category in CI dashboards.

8 Configuration Generation

8.1 Supported Platforms

Vendor	Platform ID	Compiler Class	Protocols
Cisco	ios	IOSCompiler	OSPF, BGP, ISIS, MPLS
Cisco	iosxr	IOSXRCompiler	OSPF, BGP, ISIS, SR, BFD
Cisco	nxos	NXOSCompiler	OSPF, BGP, EVPN, VXLAN
Juniper	junos	JunOSCompiler	OSPF, BGP, ISIS, MPLS, SR
Arista	eos	EOSCompiler	OSPF, BGP, EVPN, VXLAN
Nokia	sros	SROSCompiler	OSPF, BGP, ISIS, SR, L3VPN
Nokia	srlinux	SRLinuxCompiler	OSPF, BGP, EVPN
FRRouting	frr	FRRCompiler	OSPF, BGP, ISIS
SONiC	sonic	SONiCCompiler	BGP, EVPN
VyOS	vyos	VyOSCompiler	OSPF, BGP
Cumulus	cumulus	CumulusCompiler	OSPF, BGP, EVPN

8.2 Compiler Usage

```
from autonetkit.compilers.registry import get_compiler, list_platforms

# Discover available platforms
print(list_platforms())

# Compile a single device
compiler = get_compiler("iosxr")
config = compiler.compile(topo, device_id=core1.id)
cli_text = compiler.render_config(config)

# Compile all devices, routing by platform field
for router in topo.query.nodes().of_type(Router).models():
    c = get_compiler(router.data.platform)
    txt = c.render_config(c.compile(topo, router.id))
    print(f"=== {router.data.label} ===\n{txt}")
```

8.3 Jinja2 Template Customisation

Each compiler resolves templates from a search path. Drop overrides into a local templates/<platform>/ directory and set the search path:

```
compiler = get_compiler("ios")
compiler.add_template_dir("templates/ios")
# Local templates take precedence over built-in ones
```

Templates receive the compiled config object as context. Variables available in all templates:

```
# Available in every Jinja2 template:
hostname:      "Core1"
platform:      "cisco_ios"
interfaces:    [ {label: "Eth0/0", ip: "10.0.0.1/30", speed: "10G"} ]
ospf_areas:    [ {area: 0, interfaces: ["Eth0/0"]} ]
bgp_neighbors: [ {peer_ip: "10.0.0.2", remote_asn: 65002} ]
```

Caution

Not every compiler covers every protocol. If a router's topology data includes BGP configuration but the target compiler does not have a BGP template, the rendered configuration will silently omit BGP. Run the `operational.platform_support` design rule before compiling to catch coverage gaps early.

8.4 Deployment Environment Export

```
# Containerlab (container-based emulation)
from autonetkit.export import ContainerlabExporter
ContainerlabExporter().export(topo, output_dir="lab/")
# Produces: lab/topology.clab.yml

# Ansible (configuration management)
from autonetkit.export import AnsibleExporter
AnsibleExporter().export(topo, output_dir="ansible/")
# Produces: ansible/inventory.yml + playbooks/

# netsim / netlab (network simulation)
from autonetkit.export import NetsimExporter
NetsimExporter().export(topo, output_dir="sim/")
# Produces: sim/topology.yml
```

9 batteries_included Topologies

The `autonetkit.batteries_included` module ships four ready-made topology factories for common use cases. Each factory returns a fully populated `Topology` with typed models, physical links, and derived protocol layers.

9.1 Available Factories

```
from autonetkit.batteries_included import (
    datacenter_topology,
    isp_topology,
    campus_topology,
    wan_topology,
)
```

9.1.1 Leaf-Spine Datacenter

```
topo = datacenter_topology(
    spines=2,      # spine router count
    leaves=4,      # leaf switch count
    asn=65100,
    platform="arista_eos",
)
# Layers: physical, evpn (full mesh between leaves via spines)
```

9.1.2 ISP Internet Exchange

```
topo = isp_topology(
    cores=2,
    edges=4,
    route_servers=1,
    asn_base=65000, # ASN assigned sequentially
    platform="cisco_iosxr",
)
# Layers: physical, ospf (same-ASN), ebgp (cross-ASN), ibgp (full mesh)
```

9.1.3 Enterprise Campus

```
topo = campus_topology(
    core_switches=2,
    distribution_switches=4,
    access_switches=8,
    platform="cisco_ios",
)
# Layers: physical, stp, ospf
```

9.1.4 WAN Mesh

```
topo = wan_topology(
```

```

nodes=8,
seed=42,          # deterministic layout
topology="partial_mesh", # or "ring", "hub_spoke"
platform="junos",
)
# Layers: physical, ospf, ibgp (full mesh)

```

Tip

All battery-included factories accept a seed parameter for deterministic topology generation. Use the same seed in CI to ensure reproducible test scenarios.

9.2 Extending a Factory Topology

Factory topologies are ordinary Topology objects; you can add nodes, edges, or new layers after construction:

```

topo = isp_topology(cores=2, edges=4, asn_base=65000)

# Add a route reflector
rr = Router(layer="physical",
            data=RouterData(label="RR1", role="rr", asn=65000))
topo.nodes.add([rr])

# Re-derive the iBGP layer to include the new node
topo.layers.build({"ibgp": {"parent": "physical",
                           "copy_edges": False,
                           "group_by": "asn",
                           "create": "full_mesh"}})

```

10 Troubleshooting

Problem 1: NTE compilation error: NteSerialiseError: invalid NDJSON

Cause: A Pydantic model contains a field type that the NDJSON codec does not support (e.g., a nested model with circular references, or a custom type without a registered serialiser).

Solution: Add a model_serializer to the offending Pydantic class, or convert the field to a JSON-safe type (str, int, dict). Run model.model_dump_json() outside of AutoNetKit first to verify the model serialises correctly.

Problem 2: Slow performance when adding thousands of nodes one by one

Cause: Each individual topo.nodes.add([node]) call crosses the Python–Rust boundary via NDJSON serialisation. For large topologies this overhead dominates.

Solution: Batch all nodes into a single list and call topo.nodes.add(all_nodes) once. Batch insertion serialises the entire list in a single NDJSON stream, reducing round-trips by an order of magnitude. Similarly, batch-add edges after all nodes are inserted.

Problem 3: AttributeError: 'NoneType' object has no attribute 'id' when reading a node ID immediately after creating it

Cause: The node ID is assigned by NTE *after* topo.nodes.add() is called. Capturing node.id before the node has been added returns None.

Solution: Call topo.nodes.add([node]) before reading node.id. For batch adds, node IDs are set in place on all objects in the list by the time add() returns.

Problem 4: Layer derivation does not reflect nodes added after `layers.build()`

Cause: Layer derivation is not reactive. Nodes added to the parent layer after `build()` is called are not automatically cloned to derived layers.

Solution: Re-run `topo.layers.build(LAYERS)` after any structural change to the parent layer. Freeze the parent layer before building to prevent accidental post-build mutations.

Problem 5: Generated configuration is empty or missing protocol sections

Cause: The compiler for the selected platform does not have a Jinja2 template for the required protocol (e.g., no IS-IS template for `cisco_ios`).

Solution: Run the `operational.platform_support` design rule to list missing template coverage. Either add a custom template under `templates/<platform>/` or switch to a platform compiler that supports the required protocol. See Section 8 for the per-platform protocol coverage table.

Problem 6: `LayerValidationError: unknown parent layer 'physical' when calling layers.build()`

Cause: The `LAYERS` dictionary references "physical" as a parent, but no nodes have been added to a layer named "physical" yet.

Solution: Add nodes with `layer="physical"` before calling `layers.build()`. Alternatively, if using the whiteboard workflow, add a `"physical": {"parent": "whiteboard"}` entry to the `LAYERS` dictionary to derive the physical layer first.

11 Integration with the Ecosystem

AutoNetKit sits between the network engineer and a broader automation ecosystem. The following tools interoperate directly.

11.1 NTE – Graph Storage and Queries

NTE (`ank_nte`) is the Rust backend that AutoNetKit delegates to for all graph storage and columnar queries. It is installed automatically as a dependency; you do not need to interact with it directly.

For advanced use cases (custom Polars queries, direct graph traversal, petgraph algorithms), import the NTE handle from the topology store:

```
nte_handle = topo._store.nte # Rust NTE object (PyO3)
df = nte_handle.node_dataframe(layer="physical") # pl.DataFrame
```

See also: *NTE Technical Reference* (NTE-TR-001) for engine internals: graph storage (`petgraph::StableDiGraph`), Polars columnar tables, and PyO3 binding details.

11.2 NetCfg – Rust-Native Configuration Compiler

NetCfg is an alternative, high-performance configuration compiler written in Rust. Export from AutoNetKit and pass to NetCfg:

```
python -c "
import autonetkit; t = autonetkit.io.load('topology.ndjson')
autonetkit.io.export_json(t, 'topology.json')
"
netcfg compile topology.json --platform junos --output out/
```

See also: *NetCfg User Manual* (NETCFG-UM-001) for compilation targets, platform coverage, and template authoring.

11.3 TopoGen — Seed-Deterministic Topology Generation

TopoGen generates large, parameterised topologies that AutoNetKit can ingest as a starting point for design validation and config generation:

```
from topo import generate, TopologySpec
from autonetkit.io import from_topogen

spec = TopologySpec(nodes=128, seed=42, topology="partial_mesh")
gen = generate(spec)
topo = from_topogen(gen)  # Returns a populated Topology object
```

See also: *TopoGen User Manual* (TOPOGEN-UM-001) for seed parameters, topology templates, and large-scale generation strategies.

11.4 netsim / netlab — Network Simulation

AutoNetKit exports Containerlab and netsim topology files directly. Use the `NetsimExporter` to generate simulation input and then run netlab to instantiate the lab:

```
python -c "
from autonetkit.batteries_included import isp_topology
from autonetkit.export import NetsimExporter
t = isp_topology(cores=2, edges=4, asn_base=65000)
NetsimExporter().export(t, 'sim/')
"
cd sim && netlab up
```

See also: *netsim User Manual* for node image configuration and supported platform capabilities in simulation.

11.5 NetVis — Topology Visualisation

AutoNetKit exports GraphML for immediate ingestion by NetVis:

```
topo.io.to_graphml("topology.graphml")
# Open in NetVis or: netvis render topology.graphml
```

See also: *NetVis User Manual* (NETVIS-UM-001) for layout algorithms, layer overlays, and export formats.

11.6 Workbench — Orchestration

Workbench is the orchestration layer that drives the full design-to-deployment pipeline: it calls AutoNetKit for modelling, NetCfg for compilation, and netsim or Ansible for deployment. Individual AutoNetKit scripts can be wrapped as Workbench tasks:

```
# workbench.yml
tasks:
  - name: build_topology
    type: autonetkit
    script: build_isp.py
    output: topology.ndjson

  - name: compile_configs
    type: netcfg
    input: topology.ndjson
    platform: iosxr
    output: configs/
```

See also: *Workbench User Manual* (WB-UM-001) for task graph configuration, secrets management, and CI/CD integration.

12 Further Reading

- **ANK-TR-001** — *AutoNetKit: A Type-Safe Framework for Network Topology Modelling*. Architecture deep-dive: generic Pydantic data model, manager-first façade, layer mechanics, query expression engine, rule composition operators, and the configuration generation pipeline. Required reading before extending AutoNetKit with custom managers or compilers.
- **NTE-TR-001** — *NTE: Network Topology Engine Technical Reference*. Rust engine internals: petgraph::StableDiGraph storage, Polars columnar property tables, NDJSON serialisation protocol, and PyO3 binding design.
- **NETCFG-UM-001** — *NetCfg User Manual*. Rust-native configuration compiler: platform registry, Jinja2 template authoring, and batch compilation strategies.
- **TOPOGEN-UM-001** — *TopoGen User Manual*. Seed- deterministic topology generation: parameter reference and integration with AutoNetKit.
- **NETVIS-UM-001** — *NetVis User Manual*. Topology visualisation: GraphML import, interactive layer overlays, and SVG/PNG export.
- **WB-UM-001** — *Workbench User Manual*. End-to-end orchestration: task graphs, AutoNetKit integration, and deployment pipelines.